

Multi-Needle Particle Implantation Computer Assisted Surgery Based on Virtual Reality

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ABSTRACT

Multi-needle radioactive seed implantation is a scientific and effective minimally invasive technique for tumor treatment. Because of the difficulties in planning, low accuracy of puncture operation and non-standard system, it can not be widely used. In this work, a virtual reality based multi-needle particle implantation surgery planning system is developed to help doctors conduct surgical training and planning, and three-dimensional surgical guide model is constructed to assist doctors in positioning in real surgery. The results show that this work can effectively solve the problems of "unclear insertion" and "inaccurate insertion" in clinical multi-needle particle implantation.

Keywords: multi-needle particle implantation surgery; virtual surgical assistance system; organ segmentation; volume rendering; 3D reconstruction.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality.

1 INTRODUCTION

Cancer patients are on the rise globally, but cancer diagnosis and mortality rates remain high and treatment progress is slow. Multi-needle seed implantation is one of the important methods for the treatment of malignant tumors. The technique of seed implantation is called "radioactive seed implantation therapy". Because it is a minimally invasive treatment, it does not need resection and has few side effects. It has little damage and radiation, and can eliminate malignant tumors safely and quickly. In clinical statistics, the local control rate of the primary lesion can reach more than 90%, which has a good clinical effect. Due to the differences in the ability level of doctors, there are shortcomings such as insufficient understanding of the site structure of malignant tumors and inadequate surgical planning, which will lead to the problems of long operation time and high risk. To address those problems, we propose a design and method of multi-needle particle implantation surgery system based on VR. In this work, the segmentation of important human tissue based on the DeepInsight system, the visualization method of human body model based on volume rendering and the 3D reconstruction method of human skin based on point cloud data are studied. On the basis of the above key technologies, a multi-needle particle implantation surgery planning system is constructed to assist doctors to design more reasonable operation plan and reduces the risk of clinical surgery.

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2 RELATED WORK

With the continuous development of computer aided diagnosis and treatment technology and virtual surgical hardware and software equipment, virtual surgical assistant system has been used more and more widely in clinical medicine. Europe and America are pioneers in the field of virtual surgery. In 1980s, Delp et al. developed a simulation system for leg joint transplantation^[1], which can view the process of joint transplantation surgery. In 1991, Satava et al. developed a virtual abdominal surgery system^[2]. This system has the function of observing the lesion tissue and supervising the surgical process. In 2001, Karlsruhe from Germany developed the training system of laparoscopic minimally invasive surgery^[3], which used the manipulator to simulate the control of the scalpel for virtual endoscopic surgery, and realized force feedback, showing the phenomenon of human tissue cutting and deformation. In 2009, Vankipuram et al. in the United States developed a skeletal virtual surgery training system^[4]. In 2010, INRIA Institute in France developed a virtual surgery system for liver cutting^[5], which has good force feedback and simulates soft tissue realistically. In 2017, Changzheng Hospital affiliated to the Second Military Hospital developed a spinal surgery training system^[6], which simulates surgery by wearing an immersive helmet to improve doctors' surgical skills and achieve force feedback. At present, virtual surgical systems are mostly used for surgical assistance or simulation operations, while there are few studies on virtual surgical assistance systems for multi-needle particle implantation. Based on the above situation, the visualization method of human body model and three-dimensional reconstruction method of skin were studied in this paper, and the multi-needle particle implantation surgery planning system was constructed to help doctors carry out surgical training and surgical program planning.

3 METHODS

This chapter mainly introduces the design process of Multi-Needle Particle Implantation Surgery Based on VR. Firstly, the segmentation method of important human tissue based on DeepInsight is introduced. Then the visualization method of human model based on volume rendering is introduced. In order to enhance the authenticity of the model, the 3D reconstruction method of human skin based on point cloud data is introduced. Finally, the design and implementation of multi-needle seed implantation planning system are introduced.

The human body contains all kinds of important tissues. When realizing the visualization of the human body model, it is also necessary to realize the visualization of the important internal organization. This requires the segmentation of the body's important tissues to form body data, and the labeling of different tissues. Then the human body model and the important tissues of the human body are visualized by volume rendering. The DeepInsight system is used to segment important human tissues^[7,8], which is a lung intelligent assisted surgical system,

which realizes the segmentation and visualization of tissues and organs by using CT or CTA images. In Figure 1, the segmentation results of pulmonary lobes and pulmonary arteries and veins using DeepInsight system are illustrated.

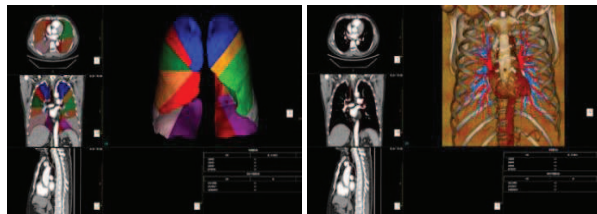


Figure 1: The segmentation of important human tissues

The visualization of human models in virtual scenes is based on the ray-casting algorithm in volume rendering, and the visualization rendering is realized by Shader in Unity 3D. The light is sampled as it passes through the volume data composed of CT data, then color and opacity are fused, and finally the 3D human body is visualized on the imaging plane. Based on the ray projection algorithm, the visualization of human body model is realized by reading CT data, which can show different structures of human body according to the adjustment of the photometric resistance. Section visualization can also be performed, showing the coronal, sagittal, and cross sections of the three-dimensional model, as well as the presentation of arbitrary sections. This is shown in Figure 2. On the basis of the ray projection algorithm, the marked parts of the human body are displayed according to the Mark file corresponding to the CT data. According to the marked numbers, the current position of the human body can be judged, and then different coloring protocols can be used to display the bronchi, arteriovenous, lung and other parts of the human body. This is shown in Figure 2. Based on the study of static human model visualization, according to 4DCT images composed of CT data of different phases, the visualization of human model under dynamic physiological environment is shown, which can simulate the changes of human body in the breathing process, mainly through the changes of lung parenchyma to show the changes of human body.

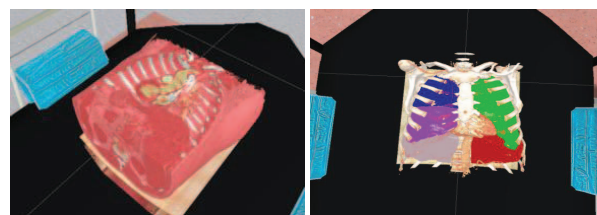


Figure 2: Coronal plane of the mannequin and Lung labeling

Enhance the realism of the mannequin by creating skin from the point cloud. Firstly, the dual synchronous depth camera was used to shoot the model from different angles to obtain the point cloud data of the model in different directions. Then, point cloud filtering denoising and normal vector estimation were used for pre-processing, and Colored ICP algorithm was used for point cloud registration. The registered point cloud data was used to construct the human skin 3D surface model by Poisson surface reconstruction algorithm. This is shown in Figure 3.

This system is used to simulate, train and plan multi-needle particle implantation operation in virtual operating room. After the surgical plan is completed, automatic puncture can be performed according to the surgical plan planned before surgery, and puncture needle will be performed according to the established plan. After the simulation of multi-needle particle

implantation, the mesh model of the surgical guide plate is generated according to the position and Angle of the puncture needle and skin data, and the surgical guide plate is obtained through 3D printing for surgical assistance. This is shown in Figure 3.

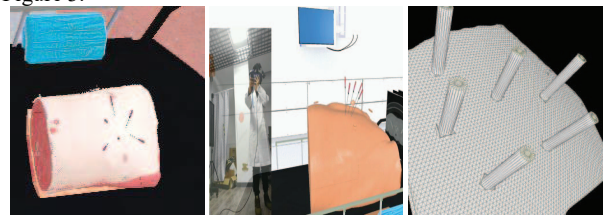


Figure 3: Human skin model and surgical template creation

4 CONCLUSION

In this paper, a multi-needle particle implantation system is designed and implemented for surgical training and assistance. We used another of our research results, DeepInsight system, to achieve organizational segmentation. Then, the visualization method of human body model based on volume rendering is studied. A 3D reconstruction method of human skin based on point cloud data is studied. In the future, it will experiment with hospitals to obtain quantitative evaluation indicators. For technical problems, volume rendering can be combined with GPU. By calculating the change gradient, the visualization state of the model at each moment is calculated to improve the coherence of.

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REFERENCES

- [1] Delp S L, Loan J P, Hoy M G, et al. An interactive graphics-based model of the lower extremity to study orthopaedic surgical procedures[J]. IEEE Transactions on Biomedical engineering, 1990, 37(8): 757-767.
- [2] Rm R M S. Virtual reality surgical simulator: The first steps[J]. Surgical Endoscopy, 1993, 7(3): 203-5.
- [3] Kühnapfel U, Cakmak H K, Maaß H. Endoscopic surgery training using virtual reality and deformable tissue simulation[J]. Computers & graphics, 2000, 24(5): 671-682.
- [4] Vankipuram M, Kahol K, McLaren A, et al. A virtual reality simulator for orthopedic basic skills: a design and validation study[J]. Journal of biomedical informatics, 2010, 43(5): 661-668.
- [5] Courtecuisse H, Jung H, Allard J, et al. GPU-based real-time soft tissue deformation with cutting and haptic feedback[J]. Progress in biophysics and molecular biology, 2010, 103(2-3): 159-168.
- [6] Pan Xianwei. Research on spinal surgery training system based on Force feedback and Immersive Virtual Reality technology [D]. Shanghai: Second Military Medical University, 2017.
- [7] Tan W, Huang P, Li X, et al. Analysis of segmentation of lung parenchyma based on deep learning methods[J]. Journal of X-ray science and technology, 2021, 29(6): 945-959.
- [8] Tan W, Liu P, Li X, et al. Segmentation of lung airways based on deep learning methods[J]. IET Image Processing, 2022, 16(5): 1444-1456.